Environmental Monitoring System Using IOT and Machine Learning Models

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Abstract— *Environmental factors such as air quality, PM2.5 levels, temperature, and humidity are all critical factors for public health, especially in Urban cities. The problem is that traditional monitoring stations are expensive and to spread out to provide good local data. To bridge this gap, our paper presents a low-cost, real-time IoT system for monitoring and forecasting environmental conditions. We made a system that uses simple IoT sensors to collect data in Real time. This data is then sent to a cloud platform where we can store the data, visualize it, and run analysis. We then use machine learning models on this data to forecast pollution levels, detect anomalies, and find patterns. Our results are that the system works successfully and turns raw sensor data into useful insights. This real-time and predictive information can help communities by spreading awareness and support better decision-making by governments. And it contributes in building smarter and more sustainable cities.*

Keywords— Air Quality, PM2.5, Temperature, Humidity, Public Health, Urban Sustainability, IoT Sensors, Real-Time Data, Cloud Platform, Machine Learning, Pollution Forecasting, Anomaly Detection, Data Visualization, Smart Cities, Predictive Analysis

I. Introduction

Urban cities are finding it harder to manage environmental health. Problems like poor air quality, rising temperatures, and unpredictable humidity are getting worse[1][2]. The main issue is that the traditional monitoring systems we use are very expensive and spread too far apart. This means they can't give us the detailed local insights we need to act quickly on public health issues[1][4]. While recent studies show that real-time data is important for greener cities and public awareness[1][2], most existing systems are centralized and costly. They also rarely include predictive analytics or make the data easy for the public to access[4][6].

This has created a clear demand for environmental monitoring solutions that are affordable, can be scaled up, and use smart technology. Cities, health officials, and infrastructure developers are all looking for cheaper alternatives that can provide real-time data, predict future trends, and be deployed easily.

Our paper introduces a system that does just that: a

Real-Time IoT-Based Environmental Monitor and Forecaster. We use low-cost sensors to gather data on air quality, temperature, and humidity. Then this data is sent to a cloud platform, where it is stored for analysis and displayed on a dashboard. We also apply machine learning models to this data to forecast pollution, spot anomalies, and uncover hidden trends. The goal is to give both communities and decision-makers the practical insights they need to build more sustainable and healthier urban environments.

II. Literature Survey

A. News and Reports: -

Nearly 99% of people worldwide breathe air that is higher than WHO guideline levels, according to the World Health Organization (WHO, 2023). Given that metropolitan areas in India often report dangerous AQI readings, particularly during the winter and festival seasons, numerous stories from The Hindu (2023) and Times of India (2024) highlight the urgent need for targeted pollution monitoring systems.

B. Books, Videos, and Courses: -

"AI for Earth Systems" (edX) highlight the application of sensor networks and AI for sustainability challenges.

C. *Market Survey: -*

The book Internet of Things: A Hands-On Approach by Arshdeep Bahga and Vijay Madisetti [3] provides insights into IoT system design, data acquisition, and cloud integration. Courses such as "IoT for Environmental Monitoring" (Coursera) and Markets and Markets (2024) projects the environmental monitoring market will reach USD 26.7 billion by 2028, with IoT-enabled devices contributing significantly [4]. However, commercial devices (e.g., Aeroqual, Kaiterra, AirVisual Pro) remain expensive, limiting adoption in developing countries.

D. Android Apps on Play Store: -

Several apps exist for environmental monitoring:

- AirVisual (IQAir) – Provides real-time AQI but relies on fixed stations.
- Plume Labs: Air Report – Offers global forecasts using community data.
- BreezoMeter – Uses ML and data fusion but with limited free features.

These apps reveal user demand but highlight limitations in affordability and transparency.

E. Research Papers: -

- Sahu et al. (2022), "IoT based air pollution monitoring system using Arduino" [5] – Proposed a low-cost prototype using MQ sensors.
- Zhou et al. (2021), "Deep learning for air quality forecasting: A survey" (Atmospheric Environment) [6] – Found LSTM/CNN most effective for AQI prediction.
- Alvear et al. (2020), "Smart cities: A survey on machine learning applications" (Sustainable Cities and Society) [7].
- Saini et al. (2021), "IoT-based framework for air quality monitoring" (IEEE Access) [8].
- Mishra & Tripathi (2023), "Low-cost environmental monitoring using IoT and cloud computing" (Journal of Cleaner Production) [9].

F. Conclusion of Literature Survey: -

Literature confirms the global demand for affordable, predictive monitoring solutions. Existing systems demonstrate feasibility but lack large-scale deployment. This creates a research gap for scalable IoT-ML frameworks, which this work addresses.

III. Proposed Methodology

A. Hardware Design and Data Acquisition: -

The new system is built with a flexible design that is low-cost, easy to expand, and collects data reliably [3][5]. The **ESP32 EVDKit V1** microcontroller is the brain of our system, chosen for its faster processing capabilities, integrated Wi-Fi, and low power consumption. The sensor connected as follows:

- **DHT-22:**
  A digital sensor for measuring ambient temperature in range (-40 to 80°C) and relative humidity (0-100% RH) with an accuracy of ±0.5°C and ±2-5% RH.
- **MQ-135:**

A sensor based on semiconductor for dectect harmful gases like NH_3, NOx, alcohol, benzene, smoke, and CO_2. The analog output is calibrated and mapped as per as the code input in Esp -32 to a standardized Air Quality Index (AQI) (0-50: Good, 50-100: Moderate, etc.) It is done by proper guidance mentioned in the documentation of the sensor.

- **GP2Y1010AU0F Optical Dust Sensor:**

This is our dust concentration by detecting the light reflectance of particles in the air integrated with Esp-32. An LED and a 150-ohm resistor are integral to its operation circuit, as specified in the datasheet. The analog output voltage is converted to Particulate Matter (PM2.5) density (µg/m³) using a published formula. The components are interconnected on a breadboard via jumper wires and powered via USB, enabling fast prototyping and deployment [8][9]. *Figure 1* shows our system's circuit diagram.

B. System Architecture and Workflow: -

The overall system architecture follows a step by step IoT procedure, reflecting smooth data flow from physical sensor to cloud-based storage and analysis[6][7]. The workflow, as in *Figure 2*, is as follows:

1. Sensing Layer: All the three sensors (DHT-22, MQ-135, GP2Y1010) continuously take the reading form the environment.
2. Data Acquisition & Preprocessing Layer: The ESP32 microcontroller reads the raw sensor signals. Accordingly, it performs necessary preprocessing, including analog-to-digital conversion (ADC) for the MQ-135 and GP2Y1010 sensors, and calibration algorithms to convert raw voltage readings into meaning full insights with proper unit (e.g., °C, %RH, AQI, µg/m³).

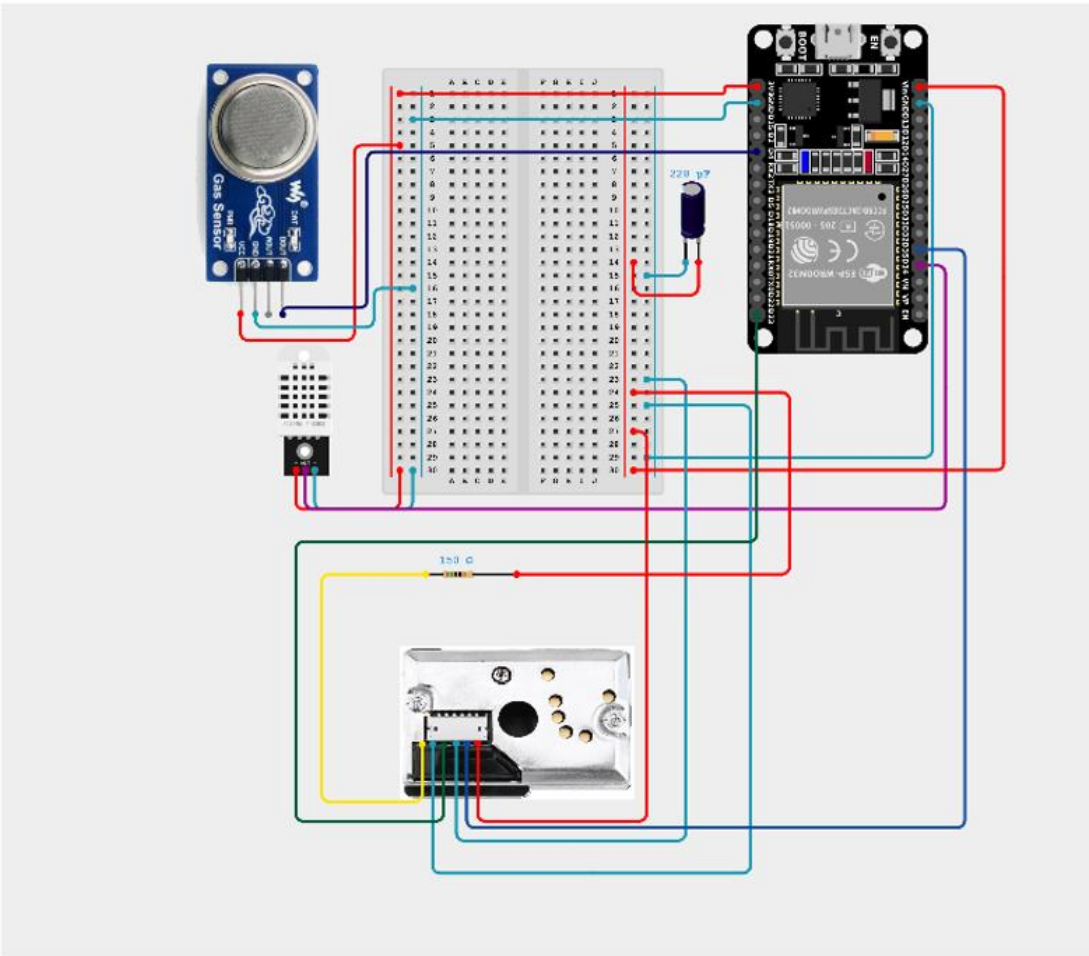
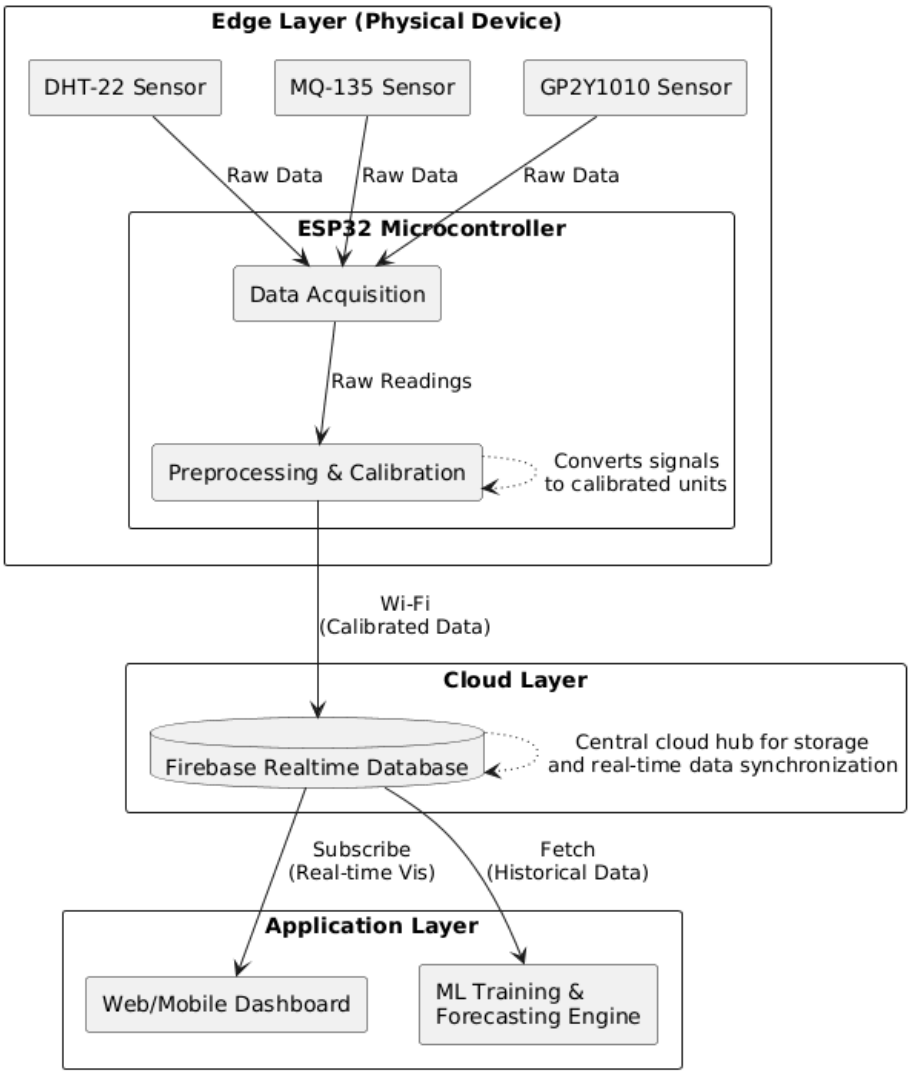


Figure 1: - Circuit Diagram of the Sensor Node

3. Data Acquisition & Preprocessing Layer: The ESP32 microcontroller reads the raw sensor signals. Accordingly, it performs necessary preprocessing, including analog-to-digital conversion (ADC) for the MQ-135 and GP2Y1010 sensors, and calibration algorithms to convert raw voltage readings into meaning full insights with proper unit (e.g., °C, %RH, AQI, µg/m³)[3][5][8].
4. Communication Layer: The processed data is timestamped (with date and time) and transmitted securely via the integrated Wi-Fi module already present in Esp-32 to the cloud layer using proper communication protocols (e.g., MQTT, HTTP REST API) [3][8][9].
5. Cloud & Storage Layer:The Google Firebase Realtime Database working as the cloud backend. It receives and stores the structured data streams from multiple nodes in real-time, providing an editable and reliable repository[8][9].
6. Application & Analytics Layer: Stored data is uses for two essential purposes:
    - Visualization: A web or mobile dashboard connects to the Firebase database to provide real-time graphical visualization of environmental parameters to end-users[9].
    - Machine Learning: Historical data is taken from Firebase for model

Figure 2: System Architecture and Data Flow



training and forecasting [6][7][8].

C. *Data Set Description: -*

For the initial proof-of-concept and model development, a proper planned data collection campaign was conducted[9].

- Collection: Using our model, we record temperature, humidity, AQI, and dust concentration readings at regular intervals (e.g., every 5 minutes).
- Duration: The data was collected continuously over a period of 4 days.
- Size: The resulting dataset consists of approximately 92 instances (4 days * 24 hours), with each instance containing the three measured features.
- Features: The dataset shows four very important numerical features: temperature, humidity and AQI, along with a timestamp for temporal analysis.
- Nature: This initial dataset is real-world but static (non-real-time for this specific training task) and will be made publicly available to foster reproducibility and further research.
- Usage for ML: The dataset is split for the train and the testing of the data. Data from the 3 days is used for training ours forecasting model. The data from the 1 day is reserved for testing and validating the model's predictive accuracy. This temporal split prevents data leakage problem and provides a realistic insight of the model's forecasting performance on unseen, future data.

IV. Experimental Setup and Methodology

A. Hardware Configuration and Data Collection: -

A breadboard was used to assemble a single monitoring node[3][5]. At 5-minute intervals, the ESP32 microcontroller was configured to read data from the DHT-22 (temperature and humidity), MQ-135(AQI-calibrated), and GP2Y1010 (PM2.5) sensors. The apparatus was placed close to a ventilation source in a fixed indoor location. For 96 hours (4days), data was continuously captured and sent in real time over Wi-Fi to a Firebase database.

B. Forecasting Methodology: -

For a temporal hold-out validation, the gathered dataset was divided into segments: The first 72 hours (3 days) of data make up the training set, and the last 24 hours (4th day) make up the test set[6]. The capacity of the model to predict future conditions based on historical trends is tested in this split[7]. Scikit-learn was used to develop five machine learning models for multi-step forward forecasting of the AQI time series:

1. A baseline linear model is called linear regression (LR).
2. Random Forest Regressor (RF): A group technique that employs one hundred decision trees
3. The Gradient Boosting Regressor (GB) is an ensemble technique that fixes faults by building trees one after the other.
4. Using five nearest neighbors, k-Nearest Neighbors (k-NN) is a non-parametric technique.
5. A model called the Support Vector Regressor (SVR) looks for the hyperplane that best fits the data.

C. Evaluation Metrics: -

Model performance was evaluated on the unseen test day using:

- **Mean Absolute Error (MAE)**
- **Root Mean Squared Error (RMSE)**
- **Coefficient of determination (R^2)**

The metrics allow comparison of absolute error (MAE), penalized large errors (RMSE), and explained variance (R^2).

V. Result and Discussion

The full set of hardware systems successfully recorded and transmitted the environmental data continuously for about 4 days without any trouble. All this data is easily accessible through the cloud Firebase system. This result ensures the system's

reliability for continuous environmental monitoring applications [8][9]. Data like temperature, humidity, and AQI is successfully fed to the machine learning baseline model to predict the AQI for 24 hours on unseen data. The best-performing model is Linear Regression with the lowest Mean Absolute Error of this greater error can be the size of the data. The least-performing model is the Support Vector Regressor with the highest MAE score but with a negative value for the R^2 score, which is evidence that it is the worst for predicting the AQI. Although AQI can be predicted based on other external factors, still our model is able to learn the trend in 7.06 and the highest R^2 score of 0.74, which means that Linear Regression can explain about 74% variance in the data. After Linear Regression, Random Forest comes 2nd in overall performance with a Mean Absolute Error equal to 8.34, which is greater than our best model. The possible cause for the low performance is small data. Overall details for all model is given in the table no.1

Fig 3 Visualization the graph of Actual AQI Verses the Predicted AQI by linear regression model. This helps to see which hour of AQI prediction is with greater error or our model predicated it worst.

Table 1: Model Performance Comparison

Model	**MAE**	**RMSE**	**R^2 Score**
Linear Regression	7.06	9.13	0.74
Random Forest	8.34	10.34	0.67
k-Nearest Neighbors	10.09	11.92	0.56
Gradient Boosting	10.37	12.11	0.55
Support Vector Regressor	16.99	19.23	-0.

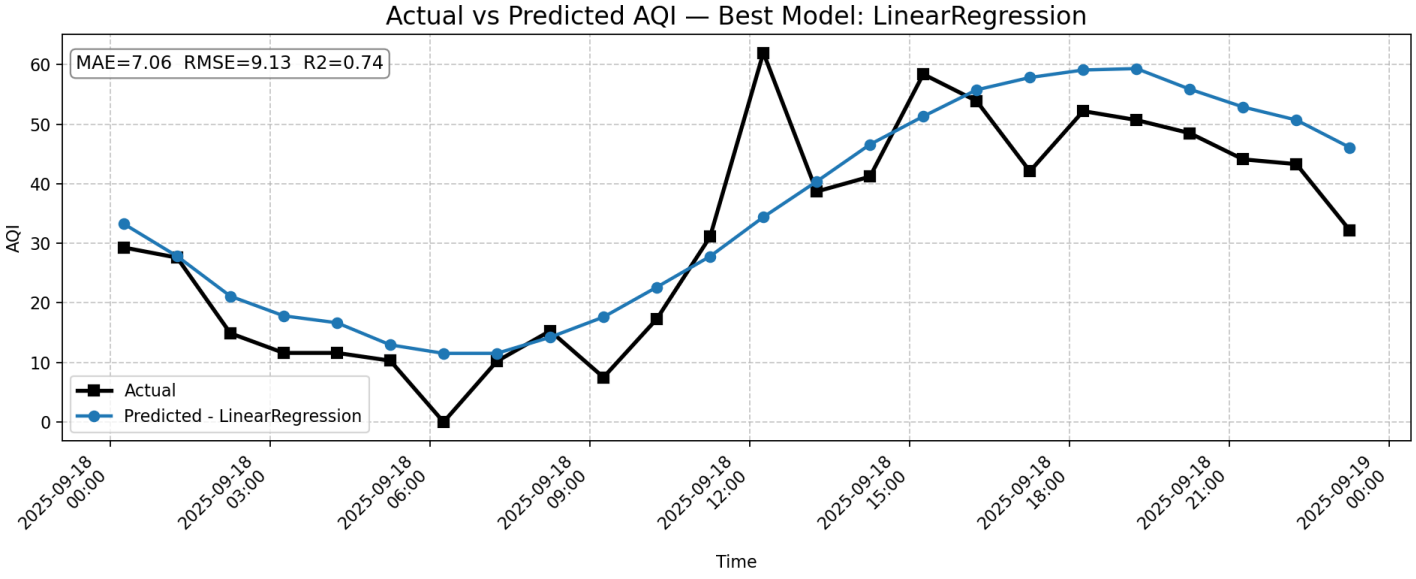


Fig 3. Actual AQI v/s Predicated AQI(Linear Regression)

VI. Conclusion

This study tells us the effectiveness of a low cost IoT-based system for real-time environmental monitoring and forecasting [1]. By integrating ESP32 microcontroller with DHT-22, MQ-135 and GP2Y1010 sensors, the system successfully captured temperature, humidity, AQI, and PM2.5 data, which was then send to a cloud platform for visualization and analysis[9]. Machine learning models were used for forecasting. Linear Regression showed the best performance among the tested machine learning models.

VII. Reference: -

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